NEWTON SCHOOL

ANSWER KEY

Multi-Step Equations & Word Problems

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Part A: Multi-Step Equations

Problem 1

$-4(x+3) = 2(x-9)$

Distributive Property | Combine Like Terms

$-4x - 12 = 2x - 18$
$-4x - 2x = -18 + 12$
$-6x = -6$
$x = 1$

$\boxed{x = 1}$

Problem 2

$5(2n-1) = -3(n+4)$

Distributive Property | Combine Like Terms

$10n - 5 = -3n - 12$
$10n + 3n = -12 + 5$
$13n = -7$
$n = -\frac{7}{13}$

$\boxed{n = -\frac{7}{13}}$

Problem 3

$7(k-2) + 5 = 3(k+1) - 8$

Distributive Property | Multi-Step

$7k - 14 + 5 = 3k + 3 - 8$
$7k - 9 = 3k - 5$
$7k - 3k = -5 + 9$
$4k = 4$
$k = 1$

$\boxed{k = 1}$

Problem 4

$-6(m-4) = -2(3m-12)$

Identity Equation

$-6m + 24 = -6m + 24$
$0 = 0$ (always true)

All real numbers

Problem 5

$8(p+2) - 3p = 5(p+4)$

No Solution

$8p + 16 - 3p = 5p + 20$
$5p + 16 = 5p + 20$
$16 = 20$ (false)

No solution

Problem 6

$-2(4y-3) = 6 - 8y$

Identity Equation

$-8y + 6 = 6 - 8y$
$6 = 6$ (always true)

All real numbers

Part A: Multi-Step Equations (continued)

Problem 7

> $3(2w+5)-4=2(w-1)+7$

Multi-Step

$6w+15-4=2w-2+7$
$6w+11=2w+5$
$6w-2w=5-11$
$4w=-6$
$w=-\frac{3}{2}$

$w=-\frac{3}{2}$ or -1.5

Problem 8

> $-5(a-2)+3a=2(5-a)$

Identity Equation

$-5a+10+3a=10-2a$
$-2a+10=10-2a$
$10=10$ (always true)

All real numbers

Part B: Word Problems

Problem 9

> The sum of three consecutive integers is 84. Find the three integers.

Consecutive Integers | Equation Setup

Let the integers be n, $n+1$, $n+2$
$n+(n+1)+(n+2)=84$
$3n+3=84$
$3n=81$
$n=27$

27, 28, 29

Problem 10

> Maria bought 5 notebooks and 3 pens for $18.50. If each pen costs $1.50, how much did each notebook cost?

Money Problem

Cost of 3 pens: $3 \times \$1.50 = \$4.50$
Cost of 5 notebooks: $\$18.50 - \$4.50 = \$14.00$
Cost per notebook: $\$14.00 \div 5 = \$2.80$

$2.80

Part B: Word Problems (continued)

Problem 11

> A rectangle has a perimeter of 56 cm. The length is 4 cm more than the width. Find the dimensions of the rectangle.

Perimeter | Geometry

Let width = w, length = $w + 4$

Perimeter: $2(w) + 2(w + 4) = 56$

$2w + 2w + 8 = 56$

$4w = 48$

$w = 12$ cm, length = 16 cm

Width: 12 cm, Length: 16 cm

Problem 12

> Jake drove 240 miles in 4 hours. For the first 2 hours, he drove at 50 mph. What was his average speed for the last 2 hours?

Rate Problem | Distance = Rate × Time

Distance in first 2 hours: $50 \times 2 = 100$ miles

Distance in last 2 hours: $240 - 100 = 140$ miles

Speed for last 2 hours: $140 \div 2 = 70$ mph

70 mph

Problem 13

> The temperature at 6 AM was -8°F. By noon, it had risen by 15°F. Then it dropped 7°F by 6 PM. What was the temperature at 6 PM?

Integer Operations | Temperature

6 AM: -8°F

Noon: $-8 + 15 = 7$°F

6 PM: $7 - 7 = 0$°F

0°F

Problem 14

> Emma has $45in5$ bills and $10 bills. She has 6 bills in total. How many of each type of bill does she have?

System of Equations | Money Problem

Let x = number of $5bills, \(y\) = numberof$10 bills

$x + y = 6$ (total bills)

$5x + 10y = 45$ (total value)

From equation 1: $x = 6 - y$

Substitute: $5(6 - y) + 10y = 45$

$30 - 5y + 10y = 45$

$5y = 15$, so $y = 3$

$x = 6 - 3 = 3$

3 five-dollar bills, 3 ten-dollar bills

Problem 15

> Twice a number decreased by 7 is equal to the number increased by 5. Find the number.

Translate to Equation

Let the number be n

$2n - 7 = n + 5$

$2n - n = 5 + 7$

$n = 12$

$n = 12$

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